Introduction in Science Unit Overview KS3

About This Unit	Lessons
This unit aims to give students an introduction into the three science subjects, including some key practical lessons. The main objective of this unit is to teach pupils how to be in safe in a laboratory and to find their way around. A great way to start your pupils' secondary science experience!	1. Health and Safety in a Lab 2. Finding Your Way Around a Lab 3. Using a Bunsen Burner 4. How to Use a Microscope 5. Physics and Investigation Skills 6. Making a Flare

1: Health and Safety in a Lab

Included in this Pack	Learning Objectives	Topics Covered	Main Pupil Activities
Lesson Presentation Teaching Ideas Chemical and Hazards Table Hazards Match-Up Activity Sheet Homework Problem-Solving Activity Sheet Safety Rules	To be safe in a laboratory and recall some common hazard symbols and laboratory rules.	To understand the health and safety issues within a laboratory. Pupils are asked to identify hazards in a lab and to recognise the common hazard symbols found on chemicals in a lab. Pupils are encouraged to list some health and safety rules to be used in the lab.	Pupils could be encouraged to complete the Hazards Match-Up Activity Sheet. Pupils could then be given the Chemicals and Hazards Table and asked to go around the room and identify hazard symbols, filling in the table. Once the rules of a lab of have been discussed, pupils could be asked complete the Safety Rules Activity Sheet. Extension: Pupils could be asked to complete the Problem-Solving Activity Sheet as an extension activity.

Differentiation	National Curriculum Aims	Resources/Practical Equipment	Health and Safety
Lower ability pupils could be given the Safety Rule Cloze Activity. Middle/higher ability pupils could be given a Problem-Solving Activity to stretch their understanding of the lesson.	Experimental Skills and Investigations • Use appropriate techniques, apparatus, and materials during fieldwork and laboratory work, paying attention to health and safety.	Examples of chemicals labelled with hazard symbols could be placed safely around the room.	Pupils should be careful when going around the room looking at the hazard symbols on the chemicals. Chemicals should be placed at intervals around the room so pupils have enough space to walk around.

2: Finding Your Way around a Laboratory

Included in this Pack	Learning Objectives	Topics Covered	Main Pupil Activities
Lesson Presentation Teaching Ideas Equipment List Equipment Match-Up Activity Sheet Homework Activity Sheet	To get to know some scientific equipment and to find your way around the laboratory.	Pupils should be able to Identify some scientific equipment found in a laboratory, both diagrams and scientific pictures. Pupils should practise drawing some common scientific equipment.	Pupils could be asked to identify some practical equipment from diagrams and drawings. They are then to be given a list of practical equipment (the Equipment List can be used here) that they need to find around the room. They are to bring back the equipment to their desk - pupils could work as a team to collect the equipment. Pupils could then choose some equipment, from the equipment already on their desk, that they would use to boil some water. They could then set up the equipment as if they were going to carry out the practical. If there is time, pupils could then be asked to draw diagrams of the equipment. Finally, to finish, pupils could do the PowerPoint quiz on identifying the scientific equipment.
Differentiation	**National Curriculum Aims**	**Resources/Practical Equipment**	**Health and Safety**
Lower ability pupils are given 3 pieces of equipment - shown on the PowerPoint. Higher ability pupils could work out all the equipment needed for boiling water. Higher ability pupils could be asked to draw their own diagram of the equipment.	Experimental Skills and Investigations • Choosing appropriate equipment for a practical. • Use appropriate techniques, apparatus, and materials during fieldwork and laboratory work, paying attention to health and safety.	Practical equipment to be placed around the room - the list can be found in the lesson pack.	Pupils to be careful when carrying the equipment back to the desk - ensure equipment is spread out around the room. If carrying out the practical (using the equipment to boil water), ensure pupils are wearing goggles, stood up, with long hair tied back and the equipment is set up properly. All bags and coats to be safely put away.

3: Using a Bunsen Burner

Included in this Pack

Lesson Presentation

Teaching Ideas

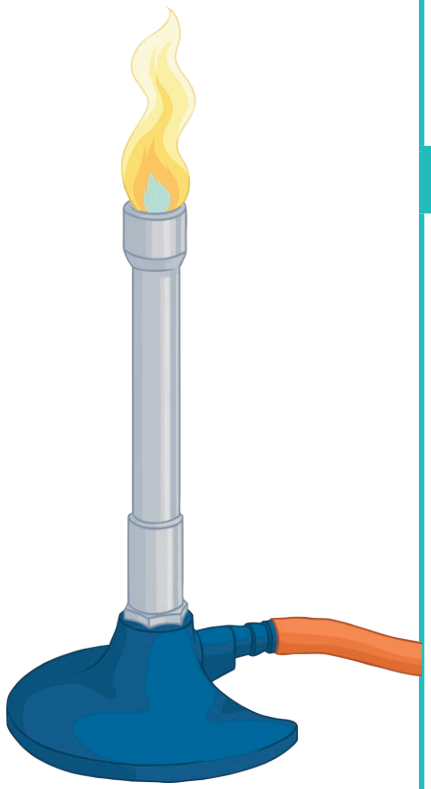
Boiling Water Activity Sheet

Bunsen Burner Licence

Bunsen Burner Activity Sheet

Bingo Sheet

Using a Bunsen Burner Practical Sheet



Learning Objectives

Be able to use a Bunsen burner safely.

Topics Covered

The parts of a Bunsen burner.

The difference between the safety flame and the roaring flame on a Bunsen burner.

How to use a Bunsen burner safely.

Main Pupil Activities

Once shown how to use a Bunsen burner, pupils could be given the **Bunsen Burner Activity Sheet** to label the Bunsen burner.

Once pupils have an understanding of how to use a Bunsen burner, they could practice changing the flame from yellow to blue and complete the **Using a Bunsen Burner Practical Sheet**.

Using a Bunsen burner, pupils then boil some water and time how long it takes to reach 100°C.

To finish and check the learning, pupils then play keyword bingo, possibly using the **Bingo Sheet**.

National Curriculum Aims

Experimental Skills and Investigations

- Choosing appropriate equipment for a practical.
- Ask questions and develop a line of enquiry based on observations of the real world, alongside prior knowledge and experience.
- Use appropriate techniques, apparatus, and materials during fieldwork and laboratory work, paying attention to health and safety.

Resources/Practical Equipment

Bunsen burner

heatproof mat

beakers

thermometers

tripod

gauze

Health and Safety

When using the Bunsen burner, pupils should be stood up, wearing goggles, and long hair tied back. Ensure all bags and coats are safely put away.

4: How to Use a Microscope

Included in this Pack

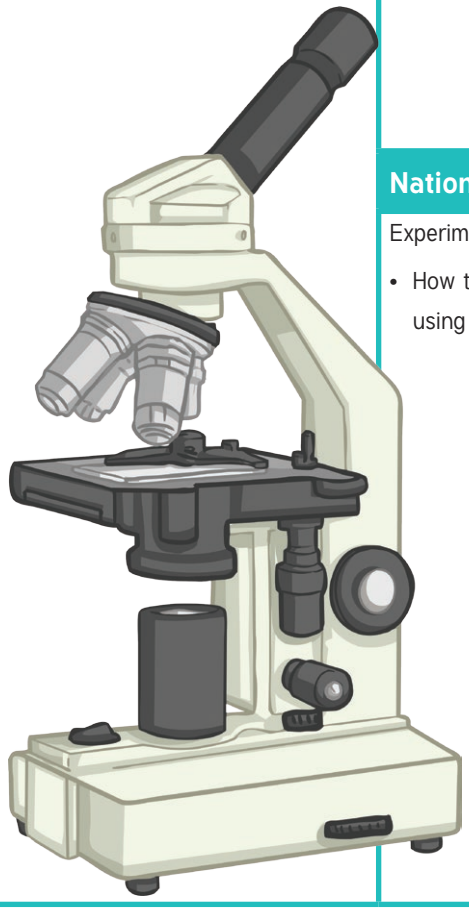
Lesson Presentation

Teaching Ideas

Microscope Observation Activity Sheet

Parts of a Microscope Activity Sheet

Parts of a Microscope Hotspot



Learning Objectives

To explore the importance of biological sciences and be able to use a microscope to magnify objects, to see in more detail.

Topics Covered

How to use a microscope to view a variety of objects, and how to focus.

Identifying the different parts of a microscope and explaining what they do.

Main Pupil Activities

Pupils could be shown how to use a microscope and what the different parts of the microscope are. At this stage, Parts of a **Microscope Hotspot activity** could be used.

They could then be given the **Parts of a Microscope Activity Sheet** to complete.

Using the microscopes, pupils could view a variety of objects and complete the **Microscope Observation Activity Sheet**.

National Curriculum Aims

Experimental Skills and Investigations

- How to observe, interpret and record structures using a light microscope.

Resources/Practical Equipment

microscopes

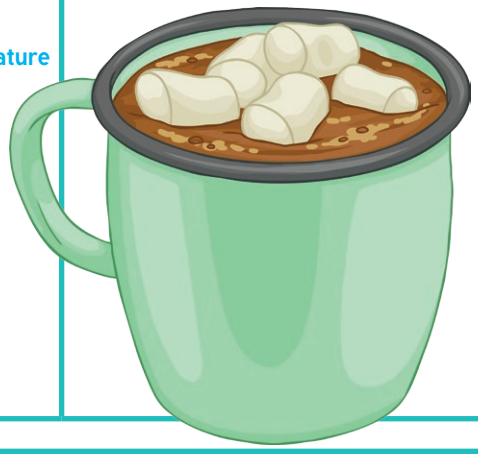
Objects to view down the microscope.

Health and Safety

Pupils are to be careful with the microscopes as the light bulb can become hot.

5: Physics and Investigation Skills

Included in this Pack	Learning Objectives	Topics Covered	Main Pupil Activities
Lesson Presentation Teaching Ideas Investigating Marshmallow Activity Sheet Results Table Activity Sheet Temperature Graph	To investigate the effect of marshmallows on the heat loss in hot chocolate.	Heat transfer and insulation. How to carry out a practical safely.	Pupils are to investigate if marshmallows keep hot chocolate warmer for longer. This practical could be set out as a full investigation, if time allows. Pupils could plot a graph of the results on Temperature Graph Sheet. This is available with pre-drawn scales for pupils that need extra support. Pupils could write a conclusion and then evaluate their experiment. If there is not time in the lesson for this, it could be set as homework.
Differentiation	**National Curriculum Aims**	**Resources/Practical Equipment**	**Health and Safety**
Higher ability students could be encouraged to explain why the variables are kept the same and to write up the practical without a prompt sheet. Middle/lower ability pupils can be given the Investigation Activity Sheet. Lower ability pupils could be given a Temperature Graph which has pre-drawn scales.	Changes in Systems • Comparing the starting with the final conditions of a system and describing increases and decreases in the amounts of energy.	beakers boiling water marshmallows hot chocolate thermometers stop clock	When carrying out the practical, pupils should be stood up, wearing goggles, with long hair tied back. Ensure all bags and coats are safely put away. Pupils to be careful with the hot water from the kettle.



6: Making a Flare

Included in this Pack

Lesson Presentation

Teaching Ideas

Making a Flare Activity Sheet

Results Table



Learning Objectives

To carry out the flame test and identify the best metal to be used in a flare.

Topics Covered

Carrying out a flame test safely.

Writing up a practical.

Main Pupil Activities

Pupils can carry out the flame test, using different compounds, to find out which one is the brightest.

They could work in pairs/small groups, taking it in turns. They can fill in the Making a Flare Activity Sheet and the Results Table.

National Curriculum Aims

Experimental Skills and Investigations

- Select, plan and carry out the most appropriate types of scientific enquiries to test predictions, including identifying independent, dependent and control variables, where appropriate.

Resources/Practical Equipment

Bunsen burner

heatproof mat

splints

metal compound

water

Health and Safety

When carrying out the practical, pupils need to be stood up, wearing goggles, with long hair tied back. Ensure all bags and coats are safely put away.